Computer Systems
Lecture 8

Overview

- Execution of instructions
- Assembly language
- Main memory, RAM
- Words, bytes and bits
- Base register
- Instruction pointer

Execution of instructions

- We have already seen the general principles of instructions execution.

- The CPU fetch-execution cycle.
- The fetch phase is the same for different types of instructions, while execute phase depends on the type of instruction.

Execution of instructions

- We will look into:
  - The types of instructions available.
  - What they can do?
  - How they can be combined to perform useful and non-trivial computations?
  - How they are used to implement HLL constructs?

Reminder: HLL and machine instruction

```
from HLL:                    i = j + k;
to assembler mnemonics:      mov EAX, [12011234]
                             add EAX, [12011238]
                             mov [1201123C], EAX

to machine binary:           0010 0000 0011 1001
                             0001 0010 0000 0001
                             0001 0010 0011 0100

                             1101 0000 1011 1001
                             0001 0010 0000 0001
                             0001 0010 0011 1000

                             0010 0011 1100 0000
                             0001 0010 0000 0001
                             0001 0010 0011 1100
```

Semantic gap

Fig. 2.3 Equivalent HLL, assembler and machine code.

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CPU and instructions

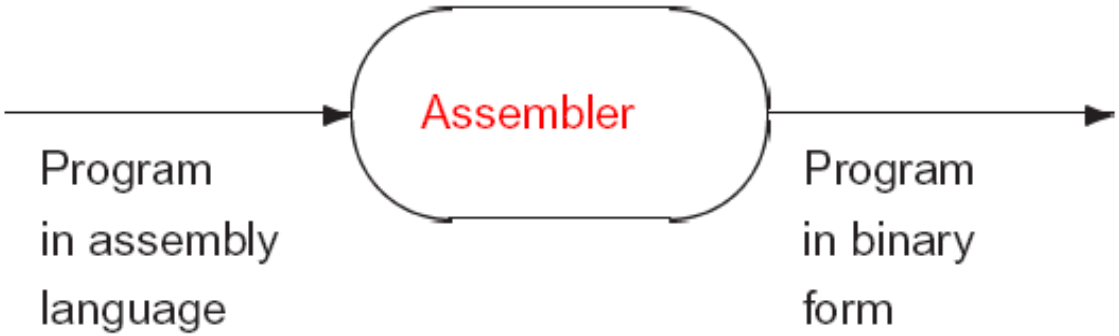
- Each different kind of CPU will be different in the set of instructions that it can perform, and in the way these are represented in binary machine code.
- General principles are similar.

- We will look into some details of **Pentium** family of CPUs.

Mnemonic form and assembly language

- In order to study more closely the way in which programs are performed, we will write some simple programs for the **Pentium** processor.
- We will write this in a mnemonic form of the Pentium machine code.
- This form is called **assembly language**.

Assembly language and assembler



- The assembler program takes as input a program written in assembly language, and:
  - Translate instructions into their binary **machine-code** form.
  - Associates labels with memory **addresses**.
  - Produces a binary **machine object code program**.

Assembly language and assembler (cont.)

- Associates **labels** with memory addresses.

```
L1:    JZ     L2
       ...
       CALL   doD
       JMP    L1
L2:    ...
```

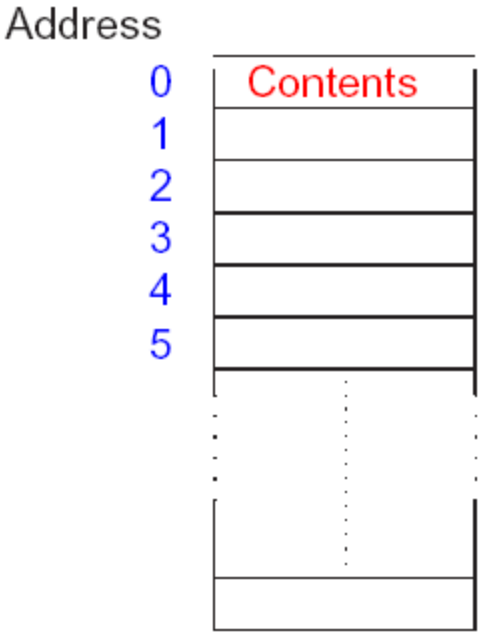
- Labels, for example, signify the beginning of a loop, the beginning of a block of memory cells, etc.
- A program has to be loaded into somewhere in main memory in order to be executed.
  - Programs must be **relocatable**.
  - What does this imply?

Assembly language and assembler (cont.)

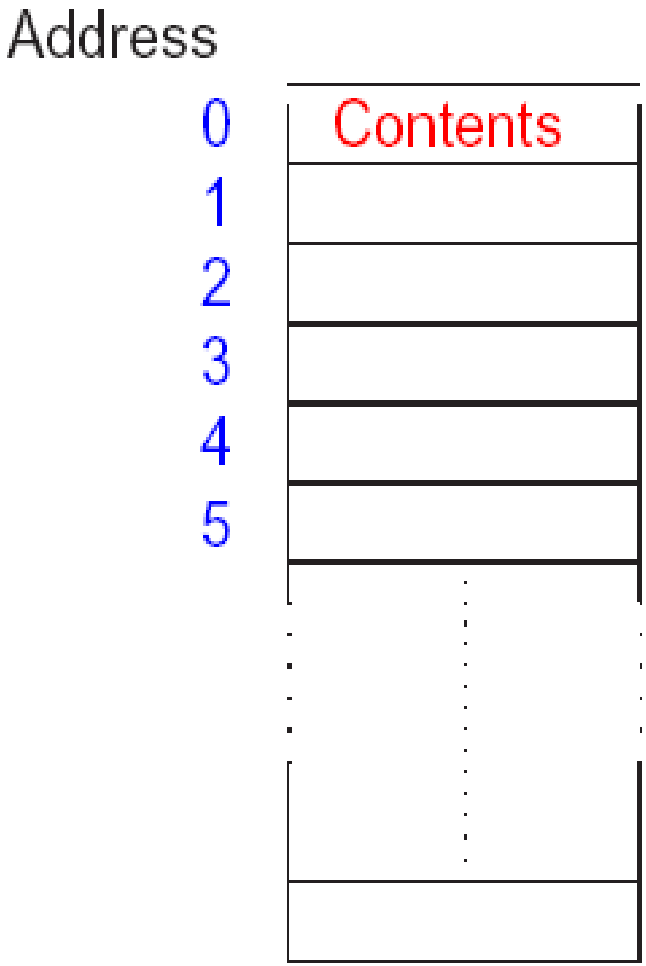
- Binary machine object code program.
  - Also called **object file**.
  - Contains binary machine code.
  - May still need a linker to generate an executable or library by linking object files together.

Main memory, RAM

- A CPU may execute only instructions loaded in the main memory (why?).
  - RAM: **Random Access Memory**.
  - The name is to set it apart from **sequential access memory** (e.g. serial tape storage) with which you cannot access data stored in the last blocks without going through the first blocks of data.



RAM



- The memory can be seen as a set of numbered storage elements, called ***words***, each of which can contain some information.
- Each word is numbered with its **address**.
- Any word of memory can be accessed "without touching" the preceding words (Random Access).
- **Access time** is the same for all the stored items.

Words, Bytes and Bits

- A **word** may be visualised as a set of more elementary storage elements (memory cells), called **bits**, arranged in a row.
- The number of bits in the word may vary between different computers; 32 bits in word is common.
- A standard unit of 8 bits called a **byte**.

CPU Registers in Pentium

- Not including the floating point registers, or the MMX (Multi-Media Extension) group.

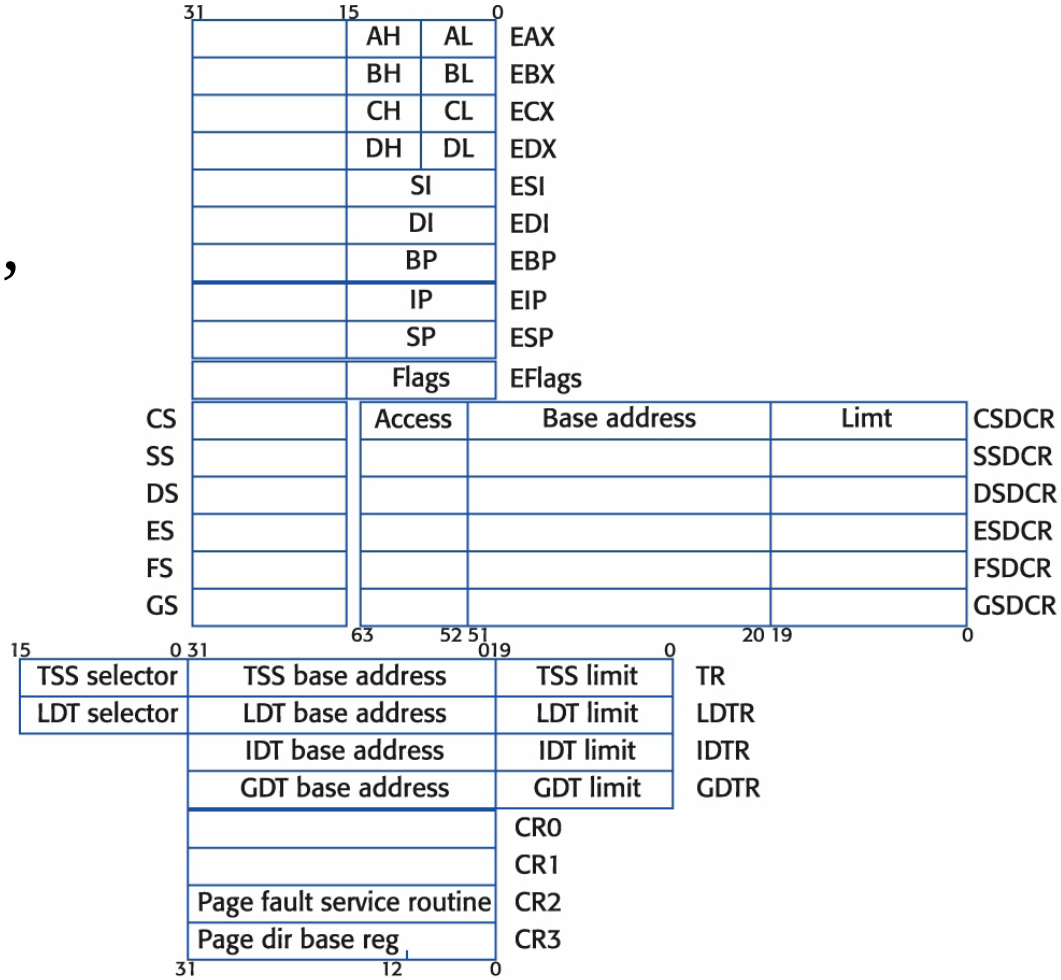


Fig. 7.5 Intel80x86/Pentium CPU register set.

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EAX register, or **A**ccumulator

- Used as a general-purpose data register during arithmetic and logical operations.
- Some instructions are optimised to work faster with the accumulator.
- In some instructions (`MUL` and `DIV`) The accumulator is assumed. No need to write it explicitly.
- It can be accessed in various ways as 8,16, or 32 bit chunks, referred to as **AL, AH, AX, EAX**.

Examples of instructions using EAX

`MOV EAX,1234H`	Load constant value 4660 into 32 bit accumulator.
`INC EAX`	Add 1 to accumulator value.
`MOV maxval,EAX`	Store accumulator value to memory variable ‘maxval’.
`DIV CX`	Divide accumulator by value in 16 bit register CX.

EBX: the Base register

- The Base register can hold addresses to point to the base of data structures, such as arrays in memory.

`LEA EBX, marks`	Initialise EBX with address of the variable ‘marks’.
`MOV AL,[EBX]`	Get 1-byte value into AL using EBX as the memory pointer.

ECX: the Count register

- It is used as a counter in loops.

`MOV ECX,100`	Initialise ECX as the FOR loop index.
……	
`for1:`	Symbolic address label.
……	
`LOOP for1`	Decrement ECX, test for zero, JMP back if non-zero.

EIP: the Instruction Pointer

- The Instruction Pointer (Program Counter) holds the address of the next instruction.

`JMP mylabel`	Forces a new address into EIP.

Further registers: ESI, EDI

- ESI: the Source Index register is used as a pointer for string or array operations.

```
MOV AX,[EBX+ESI]
```

Get one 16-bit word using Base address and Index register.

- EDI: The Destination Index register is used as a pointer for string or array operations.

```
MOV EAX,[ESI]
MOV [EDI],EAX
```

Moves one 32-bit word from source to destination.

Further registers: ESP, EBP

- EBP: The Stack Base Pointer.
- ESP: The Stack Pointer.

(more details later on)

Q&A

- The ‘execute’ phase is the same for different types of instructions. (T or F?)
- The assembler program takes as input a program written in assembly language, and translate instructions into ? form.
- Assembler associates labels with ?
- A CPU may execute only instructions loaded in the main memory (T or F ?).
- Access time is the same for all the RAM cells (T or F ?).

Q&A

- Q. What is the meaning of

  MOV EBX, [EBX] ?

  What is the side effect of this instruction?

Q&A

- Q. JNZ L2 means ?
- When will there be compile time error for the above instruction?

Q&A

- Is this valid assembly code?
- MOV maxval, loc1

Readings

- [Wil06] Section 3.6 for memory.
- [Wil06] Sections 7.1, 7.2 for the registers.